Arenula

MCP server suite for the s&box game engine

19	~120	3	8-13K
TOOLS	ACTIONS	SERVERS	TOKENS

Three Servers, One Suite

Each server has a distinct role. The Editor runs inside s&box; the API and Docs servers run standalone.

Editor

Scene manipulation, code compilation, asset management, terrain editing. Runs inside the s&box editor as a C# library plugin.

19 tools SSE :8098 C#

API

Offline type and member reference. Indexes ~1,800 types and ~15,000 members from the s&box CDN schema.

8 tools stdio TypeScript

Docs

Narrative documentation fetcher. Scrapes docs.facepunch.com, converts to Markdown, caches with LRU.

3 tools stdio TypeScript

Editor Tools

19 omnibus tools. Each takes a required `action` parameter to select the operation.

`scene`	`summary` `hierarchy` `statistics` `find` `find_in_radius` `get_details` `prefab_instances`	7
`gameobject`	`create` `destroy` `duplicate` `reparent` `rename` `enable` `set_tags` `set_transform` `batch_transform`	9
`component`	`add` `remove` `set_property` `set_enabled` `get_properties` `get_types` `copy`	7
`compile`	`trigger` `status` `errors` `generate_solution` `wait`	5

prefab	instantiate get_structure get_instances break update create save_overrides revert get_overrides	9
asset_query	browse search open get_dependencies get_model_info get_material_properties get_mesh_info get_bounds get_unsaved get_status get_json get_references	12
asset_manage	create delete rename move save reload get_references	7
editor	select get_selected set_selected clear_selection frame_selection get_play_state start_play stop_play get_log save_scene save_scene_as undo redo console_list console_run open_code_file get_preferences set_preference	18
session	list set_active load_scene	3
lighting	create configure create_skybox set_skybox	4
physics	add_collider configure_collider add_rigidbody create_model_physics create_character_controller create_joint	6
audio	create configure	2
effects	create configure_particle configure_post_processing	3
camera	create configure	2
mesh	create_block create_clutter set_face_material set_texture_params set_vertex_position set_vertex_color set_vertex_blend get_info	8
navmesh	create_agent create_area create_link generate get_status query_path	6
cloud	search get_package mount	3
project	get_collision set_collision_rule get_input get_info	4
terrain	create configure get_info paint_material sync	5

Architecture

28 C# source files organized into Core infrastructure and per-tool Handlers.

```
AI Client
    |
  HTTP / SSE
    |
ArenulaMcpServer
    |
RpcDispatcher
    |
    +-----------------+-----------------+
    |                 |                 |
Async Path        Console Path      Main Thread
compile, cloud    exception safe    all other tools
    |                 |                 |
 Handler           Handler           Handler
(background)    (sync isolation)  (GameTask.MainThread)
```

Core (8 files)

`ArenulaMcpServer` HTTP/SSE listener and session management. `RpcDispatcher` extracts tool name + action, routes to the correct handler. `ToolRegistry` holds all 19 tool schemas. `HandlerBase` provides response formatting, pagination, and error helpers. `SceneHelpers` handles scene resolution and tree walking.

Handlers (20 files)

One file per tool. Each exports a `Handle(action, args)` method with an internal switch on the action enum. Async handlers (`CompileHandler`, `CloudHandler`) use `HandleAsync` and dispatch before `GameTask.MainThread()` to prevent deadlocks.

Threading Model

Scene APIs must run on the main thread. The dispatcher calls `await GameTask.MainThread()` before invoking sync handlers. Async handlers (network/compile) run on the background thread and may internally switch to main thread for specific sub-operations.

Compared to Ozmium

Arenula is a ground-up rewrite of Ozmium MCP Server by ozmium7.

METRIC	OZMIUM	ARENULA
Tools	105	19
Total actions	105	~120
Token cost	~42,000	~8-13,000
MCP best practices	None	Full Anthropic compliance
Code compilation	Not available	trigger, status, errors, wait

METRIC	OZMIUM	ARENULA
Asset management	Read-only	Create, delete, rename, move
Prefab authoring	Instantiate only	Create, save overrides, revert
Terrain	None	Create, paint, configure, sync
Cloud assets	Deadlocked	Working (async dispatch)
Error messages	Generic	Structured with suggestions
Pagination	Truncation	offset/limit with metadata
Cross-platform	Windows P/Invoke	Clean

Design Principles

Following Anthropic's MCP tool design guidance.

Omnibus Tools

Related operations grouped with action enum parameter. Fewer tools means less selection ambiguity for the LLM.

Rich Descriptions

3-4 sentences per tool explaining what it does, when to use it, and when not to. Negative guidance prevents misuse.

Trimmed Responses

Canonical {id, name} tuples. Pagination with offset/limit. 16K character truncation. Concise and detailed modes.

Actionable Errors

Structured errors with action context and suggestions. Fuzzy type matching: "Did you mean Rigidbody?"

Schema Safety

additionalProperties: false on every schema. Enum constraints on actions and sub-types. Prevents hallucination.

Async Safety

Network tools dispatch before MainThread to avoid deadlocks. Never block the main thread with synchronous waits.

Installation

1. Copy the Editor plugin

```
# Copy into your s&box project's Libraries folder
Arenula-MCP/editor/  ⟶  YourProject/Libraries/arenula_mcp/
```

2. Build the API and Docs servers

```
# From the Arenula-MCP directory
cd api && npm install && npm run build
cd ../docs && npm install && npm run build
```

3. Configure .mcp.json in your project root

```
{
  "mcpServers": {
    "editor": { "type": "sse", "url": "http://localhost:8098/sse" },
    "api":    { "command": "node", "args": ["path/to/Arenula-MCP/api/dist/index.js"] },
    "docs":   { "command": "node", "args": ["path/to/Arenula-MCP/docs/dist/index.js"] }
  }
}
```

4. Open s&box

Arenula Editor compiles automatically and starts the MCP server on port 8098.

Based on Ozmium MCP Server by ozmium7 · API from sbox-api-mcp by sofianebel · Docs from sbox-docs-mcp
GPL-3.0 (Editor) · MIT (API, Docs)